

$C_7 \boxtimes K_4$ is Not Biplanar

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September 20, 2026

1 Introduction

In a survey of the Earth-Moon problem from 2018, Ellen Gethner conjectured that the graphs $C_5 \boxtimes K_4 - v$ and $C_7 \boxtimes K_4$ are instances of 10-chromatic biplanar graphs. If so, these would be the first known graphs with those two properties [1]. These two graphs can easily be shown to be 10-chromatic through their independence number, and thus all that remains is to check if these two graphs are biplanar. It has been proven through exhaustive search that $C_5 \boxtimes K_4 - v$ is not biplanar [2]. This paper proves that $C_7 \boxtimes K_4$ is also not biplanar.

1.1 Definitions and Preliminaries

Definition 1. Given a graph G and two disjoint sets of vertices $V_1, V_2 \subset V(G)$,

$$N_G(V_1, V_2) = \{v \in V_1 : \exists u \in V_2, (v, u) \in E(G)\}.$$

Definition 2. Given two graphs G_1, G_2 where $V(G_1) = V(G_2)$, let $G = G_1 \cup G_2$ denote the graph where $V(G) = V(G_1) = V(G_2)$ and $E(G) = E(G_1) \cup E(G_2)$.

In particular, a graph G is biplanar if and only if there exist planar graphs G_1, G_2 such that $G = G_1 \cup G_2$. We restate the following well-known lemmas.

Lemma 1. Given a planar graph G , the graph formed by contracting any number of edges in G is also planar.

Lemma 2. The graph K_9 is not biplanar. [3]

Given K_9 is not biplanar, the largest possible clique number for a biplanar graph is 8. For potentially biplanar graphs G with $\omega(G) = 8$, one strategy to show that G is not biplanar is to show that any decomposition of G into two planar graphs $G_1 \cup G_2$ would imply the existence of a decomposition of K_9 into two planar graphs. The existence of a clique of size 8 makes this task easier. We build upon this intuition with the following result.

Lemma 3. Let G be a biplanar graph with $\omega(G) = 8$ where G_1, G_2 are planar graphs such that $G = G_1 \cup G_2$ and $V' \subseteq V(G)$ is any set of 8 vertices fully connected in G . There cannot exist connected components C_1 from the induced subgraph $G_1 - V'$ and C_2 from $G_2 - V'$ such that $N_{G_1}(V', C_1) \cup N_{G_2}(V', C_2) = V'$.

Proof. Suppose $G = G_1 \cup G_2$ where G_1, G_2 are planar graphs, $\{v_1, \dots, v_8\} = V' \subseteq V(G)$ is a collection of 8 fully connected vertices in G , and suppose for the sake of contradiction there exist connected components C_1 from $G_1 - V'$ and C_2 from $G_2 - V'$ such that $N_{G_1}(V', C_1) \cup N_{G_2}(V', C_2) = V'$.

For both $i \in \{1, 2\}$, let H_i denote the graph obtained by starting with the induced subgraph $G_i[V' \cup C_i]$ and then repeatedly performing edge contractions among the vertices in the set C_i until only one vertex from this set remains, which we will label v_9 (in both H_1, H_2). It is possible to guarantee that edge contractions will yield a singular vertex v_9 in both graphs since C_i is connected in G_i . Any $v \in V'$ is connected to v_9 in H_i if and only if $v \in N_{G_i}(V', C_i)$. We can also guarantee that H_1, H_2 are planar given both are obtained by

performing edge contractions on a vertex induced subgraph of a planar graph, and we know from Lemma 1 that edge contractions preserve planarity.

H_1, H_2 are both planar graphs defined on the set of vertices $\{v_1, \dots, v_9\}$. We will show that $H_1 \cup H_2$ is fully connected. For any pair $v, u \in V'$, $(v, u) \in G$ given that the vertices in V' are fully connected in G , and as a consequence $(v, u) \in E(G_1)$ or $(v, u) \in E(G_2)$. Then, note that $(v, u) \in H_i$ if and only if $(v, u) \in G_i$ for $i \in \{1, 2\}$ given $H_i[V'] = G_i[V']$ by construction; therefore, $(v, u) \in E(H_1 \cup H_2)$. For any vertex $v \in V'$, $(v, v_9) \in E(H_1 \cup H_2)$ given $v \in V' = N_{G_1}(V', C_1) \cup N_{G_2}(V', C_2)$ and we had previously identified that v_9 is connected to $N_{G_1}(V', C_1)$ in H_1 and $N_{G_2}(V', C_2)$ in H_2 .

Given H_1, H_2 are planar graphs and $H_1 \cup H_2 = K_9$, this contradicts Lemma 2. $\square$

The following result is a simple property of the graph K_8 , which we later rely on for the main result.

Lemma 4. If G_1, G_2 are planar graphs such that $G_1 \cup G_2 = K_8$, the largest connected components in both G_1, G_2 have at least 6 vertices.

Proof. Let G_1, G_2 be planar graphs where $G_1 \cup G_2 = K_8$. Let $V' \subseteq V(G_1)$ represent the vertices of the largest connected component in G_1 (the following argument applies without loss to generality to G_2).

For the case $3 \leq |V'| < 6$, we know there are no edges in G_1 that connect any vertices in V' to $V(G) - V'$. All of these missing edges are necessarily present in G_2 , and given that V' and $V(G) - V'$ both have a size of at least 3, this implies that $K_{3,3}$ is a subgraph G_2 . This indicates that G_2 is not planar due to Kuratowski's theorem, which contradicts our initial assumption.

If $|V'| \leq 2$, G_1 consists of at most 4 disconnected edges; given $|E(K_8)| = 28$, this implies $|E(G_2)| \geq 24$. The well-known edge bound for planar graphs states that a graph on 8 vertices has at most $3 \times 8 - 6 = 18$ edges, and this contradiction completes the proof. $\square$

2 Properties of $C_7 \boxtimes K_4$

Here we focus on proving the main result regarding the graph $C_7 \boxtimes K_4$.

Lemma 5. If G_1, G_2 are planar graphs such that $G_1 \cup G_2 = C_7 \boxtimes K_4$, G_1 and G_2 are connected graphs.

Proof. For any planar graph G , given $|V(G)| \geq 3$, it is known that $|E(G)| \leq 3|V(G)| - 6$ as a consequence of the edge limit for planar graphs. As a consequence, given G_1, G_2 are planar graphs with $G_1 \cup G_2 = C_7 \boxtimes K_4$, $|E(G_1)|, |E(G_2)| \leq 3 \times 28 - 6 \leq 78$. Additionally, $|E(C_7 \boxtimes K_4)| = 154$, and so $|E(G_1)| + |E(G_2)| \geq 154$. From these relationships, we can observe that $76 \leq |E(G_1)|, |E(G_2)| \leq 78$.

For any $G_i \in \{G_1, G_2\}$, assume for the sake of contradiction $V(G_i)$ can be split into two nonempty sets V_1, V_2 such that in G_i , no vertex in V_1 is connected to a vertex in V_2 . Note that the induced subgraphs $G_i[V_1], G_i[V_2]$ are planar graphs given G_i is planar. If $|V_1|, |V_2| \geq 3$, by using the edge limit inequality for planar graphs, we observe that

$$|E(G_i)| = |E(G_i[V_1])| + |E(G_i[V_2])| \leq (3|V_1| - 6) + (3|V_2| - 6) = 3|V(C_7 \boxtimes K_4)| - 12 = 72,$$

which contradicts our previous assertion that $|E(G_i)| \geq 76$. For the case where $|V_1| = 2$, it follows that $|V_2| = 26$ and $|E(G_i[V_1])| \leq 1$, and thus

$$|E(G_i)| = |E(G_i[V_1])| + |E(G_i[V_2])| \leq 1 + (3|V_2| - 6) = 73,$$

which again contradicts our assertion that $|E(G_i)| \geq 76$; this applies without loss of generality to the case $|V_2| = 2$. Lastly, if $|V_1| = 1$, $|V_2| = 27$ and $|E(G_i[V_1])| = 0$, so $|E(G_i)| = |E(G_i[V_2])| \leq 3 \times 27 - 6 = 75$, which contradicts $|E(G_i)| \geq 76$; this applies without loss of generality to the case $|V_2| = 1$. $\square$

In the graph $C_7 \boxtimes K_4$, we will say that the 28 vertices can be divided into 7 **vertex bundles** of size 4, which we will denote as $\{V_i\}_{\mathbb{Z}/7\mathbb{Z}}$, where each vertex $v \in V_i$ is connected to all vertices in V_{i-1}, V_i, V_{i+1} excluding itself. We will rely on this representation of $C_7 \boxtimes K_4$ as a collection of vertex bundles in the following two results.

Lemma 6. Assume G_1, G_2 are planar graphs such that $G_1 \cup G_2 = C_7 \boxtimes K_4$. Given any two adjacent vertex bundles V_i, V_{i+1} , there exist vertex bundles V_{j-1}, V_j, V_{j+1} distinct from V_i, V_{i+1} where the vertices in V_j can be split into two pairs $\{v_{j,1}, v_{j,2}\}$ and $\{v_{j,3}, v_{j,4}\}$ and at least one of the following is true

1. $N_{G_1}(V_{j+1} \cup \{v_{j,1}, v_{j,2}\}, V_{j-1} \cup \{v_{j,3}, v_{j,4}\}) = \emptyset$
2. $N_{G_2}(V_{j+1} \cup \{v_{j,1}, v_{j,2}\}, V_{j-1} \cup \{v_{j,3}, v_{j,4}\}) = \emptyset$

Proof. Let G_1, G_2 denote planar graphs such that $G_1 \cup G_2 = C_7 \boxtimes K_4$. Given an arbitrary $i \in \mathbb{Z}/7\mathbb{Z}$, let $H_1 = G_1 - (V_i \cup V_{i+1})$, $H_2 = G_2 - (V_i \cup V_{i+1})$.

We first assume for the sake of contradiction that there exists a connected components C_1 in H_1 and C_2 in H_2 where $C_1 \cap V_{i-1} \neq \emptyset$, $C_2 \cap V_{i-1} \neq \emptyset$, $C_1 \cap V_{i+2} \neq \emptyset$, and $C_2 \cap V_{i+2} \neq \emptyset$. In other words, we assume not all vertices in V_{i+2} are path disconnected from every vertex in V_{i-1} within both H_1, H_2 .

Because $H_1[V_{i-1} \cup V_{i-2}] \cup H_2[V_{i-1} \cup V_{i-2}] = G[V_{i-1} \cup V_{i-2}] = K_8$, either $H_1[V_{i-1} \cup V_{i-2}]$ is a connected graph or $H_2[V_{i-1} \cup V_{i-2}]$. This is the case because if $V_{i-1} \cup V_{i-2}$ could be partitioned into two sets A, B where no vertex in A is directly connected to any vertex in B within $H_1[V_{i-1} \cup V_{i-2}]$, then all vertices in A are connected to all vertices in B within $H_2[V_{i-1} \cup V_{i-2}]$ and thus $H_2[V_{i-1} \cup V_{i-2}]$ is a connected graph. This also holds when H_1, H_2 are reversed. It follows that one of the following is true: $H_1[V_{i-1} \cup V_{i-2}]$ is connected and thus $V_{i-1} \subseteq C_1$, or $H_2[V_{i-1} \cup V_{i-2}]$ is connected and thus $V_{i-1} \subseteq C_2$. Without loss to generality assume $V_{i-1} \subseteq C_1$. Given any $v \in V_i$, $v \in N_{G_1}(V_i, C_1)$ if v is connected to any vertex in V_{i-1} in G_1 ; if v is not connected to any vertex in V_{i-1} , v is necessarily connected to every vertex in V_{i-1} in G_2 , and given $C_2 \cap V_{i-1} \neq \emptyset$, $v \in N_{G_2}(V_i, C_2)$. This shows that, $v \in N_{G_1}(V_i, C_1) \cup N_{G_2}(V_i, C_2)$ for all $v \in V_i$, and thus $V_i = N_{G_1}(V_i, C_1) \cup N_{G_2}(V_i, C_2)$. This argument can be repeated for the induced subgraphs $H_1[V_{i+2} \cup V_{i+3}]$, $H_2[V_{i+2} \cup V_{i+3}]$ to show that $V_{i+1} = N_{G_1}(V_{i+1}, C_1) \cup N_{G_2}(V_{i+1}, C_2)$. From this, we can see that $V_i \cup V_{i+1} = N_{G_1}(V_i \cup V_{i+1}, C_1) \cup N_{G_2}(V_i \cup V_{i+1}, C_2)$. Given that $V_i \cup V_{i+1}$ is a set of 8 fully connected vertices in $C_7 \boxtimes K_4$ and C_1, C_2 are connected components of $G_1 - (V_i \cup V_{i+1}), G_2 - (V_i \cup V_{i+1})$ respectively, Lemma 3 shows that G_1, G_2 cannot be planar graphs such that $G_1 \cup G_2 = C_7 \boxtimes K_4$, which contradicts our initial assumption. We conclude that V_{i-1}, V_{i+2} are path disconnected from each other in at least one of H_1, H_2 .

We now consider the case where V_{i-1}, V_{i+2} are path disconnected from each other in H_1 , and we will show that condition (1) in the claim is true. If V_{i-1}, V_{i+2} is chosen to be path disconnected to each other in H_2 instead, condition (2) can be shown to be true with the equivalent argument without loss to generality.

In the graph H_1 , let C denote the set of all vertices in the same connected component with at least one vertex in V_{i-1} . Let C_k denote the value $|V_k \cap C|$. By definition, $C_{i-1} = 4$ and $C_{i+2} = 0$. Within the sequence of vertex bundles $V_{i-1}, V_{i-2}, V_{i-3}, V_{i+3}, V_{i+2}$ that make up $V(H_1)$, for any arbitrary pair of adjacent vertex bundles V_j, V_{j+1} from this sequence, it is necessarily true that $C_j + C_{j+1} \leq 2$ or $C_j + C_{j+1} \geq 6$. This is known because the set of vertices in $C \cap (V_j \cup V_{j+1})$ is disconnected from the set $(V_j \cup V_{j+1}) - C$ in $H_1[V_j \cup V_{j+1}]$, and given $H_1[V_j \cup V_{j+1}] \cup H_2[V_j \cup V_{j+1}] = K_8$, Lemma 4 guarantees that the largest connected component in $H_1[V_j \cup V_{j+1}]$ has at least 6 vertices; this clearly cannot be the case if $|C \cap (V_j \cup V_{j+1})| < 6$ and $|(V_j \cup V_{j+1}) - C| < 6$. We thus know that the sequence of values $C_{i-1}, C_{i-2}, C_{i-3}, C_{i+3}, C_{i+2}$ must start at 4, end at 0, and the sum of every adjacent pair of values is either at least 6 or at most 2. We know $C_{i-1} + C_{i-2} \geq 6$ and $C_{i+3} + C_{i+2} \leq 2$ due to the values of C_{i-1}, C_{i+2} respectively. Let $C_j \in C_{i-2}, C_{i-3}, C_{i+3}$ denote the last value in this sequence such that $C_{j-1} + C_j \geq 6$; it follows that $C_j + C_{j+1} \leq 2$. Given $C_{j-1}, C_j, C_{j+1} \in \{0, 1, 2, 3, 4\}$, the only possible values where these inequalities hold are $C_{j-1} = 4, C_j = 2, C_{j+1} = 0$. This implies V_j has two sets $\{v_{j,1}, v_{j,2}\}$ and $\{v_{j,3}, v_{j,4}\}$ where all vertices in $V_{j-1} \cup \{v_{j,3}, v_{j,4}\}$ are in a connected component with at least one vertex in V_{i-1} within G_1 , and no vertices in $V_{j+1} \cup \{v_{j,1}, v_{j,2}\}$ are in a connected component with at a vertex in V_{i-1} . It follows that $N_{G_1}(V_{j+1} \cup \{v_{j,1}, v_{j,2}\}, V_{j-1} \cup \{v_{j,3}, v_{j,4}\}) = \emptyset$, and clearly V_{j-1}, V_j, V_{j+1} are disjoint from V_i, V_{i+1} . This completes the proof. $\square$

For any graph G defined over the set of vertices in $\{V_i\}_{\mathbb{Z}/7\mathbb{Z}}$, we will say G has a **split** at the vertex bundles V_{i-1}, V_i, V_{i+1} if the vertices in V_i can be divided into two pairs $\{v_{j,1}, v_{j,2}\}$ and $\{v_{j,3}, v_{j,4}\}$ such that $N_G(V_{i+1} \cup \{v_{j,1}, v_{j,2}\}, V_{i-1} \cup \{v_{j,3}, v_{j,4}\}) = \emptyset$. This is shorthand that will be used to more easily refer to the results of Lemma 6.

Theorem 1. $C_7 \boxtimes K_4$ is not a biplanar graph.

Proof. Assume for the sake of contradiction that G_1, G_2 are planar graphs such that $G_1 \cup G_2 = C_7 \boxtimes K_4$, where both G_1, G_2 are graphs defined on the set of vertices contained in the vertex bundles $\{V_i\}_{i \in \mathbb{Z}/7\mathbb{Z}}$.

We will first prove that there exist two disjoint sets of 3 bundles V_{i-1}, V_i, V_{i+1} and V_{j-1}, V_j, V_{j+1} where V_{i-1}, V_i, V_{i+1} is a split in G_1 or G_2 , and V_{j-1}, V_j, V_{j+1} is also a split in G_1 or G_2 . First, among the vertex bundles $\{V_i\}_{i \in \mathbb{Z}/7\mathbb{Z}}$, choose an arbitrary pair of adjacent bundles V_x, V_{x+1} . Lemma 6 shows that there exist bundles V_{i-1}, V_i, V_{i+1} (disjoint from V_x, V_{x+1}) where either G_1 or G_2 is split at V_{i-1}, V_i, V_{i+1} . Next, given bundles V_{i-1}, V_i , apply Lemma 6 again to show that there exist bundles V_{j-1}, V_j, V_{j+1} disjoint from both V_{i-1}, V_i where G_1 or G_2 is split at V_{j-1}, V_j, V_{j+1} . If V_{i+1} is also not in V_{j-1}, V_j, V_{j+1} , that completes the proof as V_{i-1}, V_i, V_{i+1} and V_{j-1}, V_j, V_{j+1} are disjoint. Otherwise, given $V_{i+1} \in \{V_{j-1}, V_j, V_{j+1}\}$ and $V_{i-1}, V_i \notin \{V_{j-1}, V_j, V_{j+1}\}$, we can infer $V_{j-1} = V_{i+1}$. Lastly, apply Lemma 6 again on V_i, V_{i+1} to produce bundles V_{k-1}, V_k, V_{k+1} which contain a split in either G_1 or G_2 . Again, if V_{k-1}, V_k, V_{k+1} does not include V_{i-1} , that completes the proof as V_{i-1}, V_i, V_{i+1} and V_{k-1}, V_k, V_{k+1} are disjoint. Otherwise, we can observe that $V_{i-1} = V_{k+1}$, and given $V_{i+1} = V_{j-1}$, we can see that V_{j-1}, V_j, V_{j+1} and V_{k-1}, V_k, V_{k+1} are disjoint.

Next, given two disjoint sets of bundles V_{i-1}, V_i, V_{i+1} and V_{j-1}, V_j, V_{j+1} that define splits in either G_1 or G_2 , we prove that exactly one set defines a split in G_1 and the other set defines a split in G_2 . Assume for the sake of contradiction that G_1 has a split at both V_{i-1}, V_i, V_{i+1} and V_{j-1}, V_j, V_{j+1} . Note that the following argument also holds without loss of generality if G_1 is replaced with G_2 . Given that there are 7 vertex bundles in total and V_{i-1}, V_i, V_{i+1} is disjoint from V_{j-1}, V_j, V_{j+1} , we can infer that $V_{i-1} = V_{j+2}$ or $V_{i+1} = V_{j-2}$; we assume without loss of generality that $V_{i-1} = V_{j+2}$. From the definition of splits, we know that V_i can be split into two sets $\{v_{i,1}, v_{i,2}\}$ and $\{v_{i,3}, v_{i,4}\}$ such that $N_{G_1}(V_{i+1} \cup \{v_{i,1}, v_{i,2}\}, V_{i-1} \cup \{v_{i,3}, v_{i,4}\}) = \emptyset$, and we know that V_{i-3} can be split into two sets $\{v_{i-3,1}, v_{i-3,2}\}$ and $\{v_{i-3,3}, v_{i-3,4}\}$ such that $N_{G_1}(V_{i-4} \cup \{v_{i-3,1}, v_{i-3,2}\}, V_{i-2} \cup \{v_{i-3,3}, v_{i-3,4}\}) = \emptyset$. Given $C_7 \boxtimes K_4$ has no edges connecting vertices from non-adjacent vertex sets, there are no edges in G_1 connecting $V_{i+1} \cup \{v_{i,1}, v_{i,2}\}$ to $V_{i-2} \cup \{v_{i-3,3}, v_{i-3,4}\}$, and there are no edges connecting $V_{i-1} \cup \{v_{i,3}, v_{i,4}\}$ to $V_{i-4} \cup \{v_{i-3,1}, v_{i-3,2}\}$. Lastly, G_1 has no edges connecting V_{i+2} directly to any of $V_i, V_{i-1}, V_{i-2}, V_{i-3}$. As a result,

$$N_{G_1}((V_{i-4} \cup \{v_{i-3,1}, v_{i-3,2}\}) \cup V_{i+2} \cup (V_{i+1} \cup \{v_{i,1}, v_{i,2}\}), (V_{i-1} \cup \{v_{i,3}, v_{i,4}\}) \cup (V_{i-2} \cup \{v_{i-3,3}, v_{i-3,4}\})) = \emptyset.$$

This demonstrates that $V(G_1)$ can be partitioned into two sets of vertices where no edge in G_1 connects any vertex in one to a vertex in the other. This contradicts Lemma 5 as this implies G_1 is not connected.

Lastly, given there exist disjoint sets of bundles V_{i-1}, V_i, V_{i+1} and V_{j-1}, V_j, V_{j+1} where one defines a split in G_1 and the other defines a split in G_2 , we complete the proof of the theorem. Without loss to generality we assume G_1 has a split in V_{i-1}, V_i, V_{i+1} and G_2 has a split in V_{j-1}, V_j, V_{j+1} . Also, we again assume without loss to generality that $V_{i-1} = V_{j+2}$. Let $H_1 = G_1 - (V_{i-1} \cup V_{i-2})$ and $H_2 = G_2 - (V_{i-1} \cup V_{i-2})$. As a consequence of the splits, there exist vertices $\{v_{i,1}, v_{i,2}\} \subseteq V_i$ such that in G_1 , there are no edges connecting any vertex in V_{i-1} to either $v_{i,1}, v_{i,2}$; this implies all vertices in V_{i-1} are connected to both $v_{i,1}, v_{i,2}$ in G_2 , and thus if C_2 is the connected component in H_2 that contains $v_{i,1}$, $V_{i-1} = N_{G_2}(V_{i-1}, C_2)$. Similarly, there exist vertices $\{v_{i-3,1}, v_{i-3,2}\} \subseteq V_{i-3}$ such that in G_2 , there are no edges connecting any vertex in V_{i-2} to either $v_{i-3,1}, v_{i-3,2}$; all vertices in V_{i-2} are connected to both $v_{i-3,1}, v_{i-3,2}$ in G_1 , and thus if C_1 is the connected component in H_1 that contains $v_{i-3,1}$, $V_{i-2} = N_{G_1}(V_{i-2}, C_1)$. This shows that $V_{i-2} \cup V_{i-1} = N_{G_1}(V_{i-2} \cup V_{i-1}, C_1) \cup N_{G_2}(V_{i-2} \cup V_{i-1}, C_2)$, and given $G_1[V_{i-2} \cup V_{i-1}] \cup G_2[V_{i-2} \cup V_{i-1}] = K_8$, Lemma 3 shows that G_1, G_2 cannot be planar graphs such that $G_1 \cup G_2 = C_7 \boxtimes K_4$, which contradicts our initial assumptions and thus completes the proof. $\square$

References

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